Industrial Internship Report on

” Smart City Traffic Forecasting”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by Upskill Campus in collaboration with UniConverge Technologies Pvt Ltd. The internship focused on solving a real-world problem using Machine Learning techniques. The project undertaken was "Smart City Traffic Forecasting", which aims to predict traffic volume based on time-related features such as hour, day, and month. During the internship, I worked on data preprocessing, feature engineering, and implementing machine learning models such as Linear Regression and Random Forest. These models were used to analyze traffic patterns and predict future traffic volume. This internship gave me practical exposure to real-world data science problems and helped me improve my programming and analytical skills. It was an overall valuable learning experience.

Acknowledgement

I would like to express my sincere gratitude to Upskill Campus and UniConverge Technologies Pvt Ltd (UCT) for providing me with the opportunity to undertake this industrial internship.

I am thankful to my mentors and instructors for their valuable guidance, support, and encouragement throughout the internship. Their assistance helped me understand the concepts of Data Science and Machine Learning and successfully complete my project.

I would also like to thank my college and faculty members for their continuous support and motivation. I am grateful to my friends and peers who helped me during this internship.

This internship has been a great learning experience, and I am thankful to everyone who contributed directly or indirectly to my successful completion of this project.

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1 Preface

This internship provided me with a valuable opportunity to gain practical exposure in the field of Data Science and Machine Learning. The project assigned to me was “Smart City Traffic Forecasting”, which focuses on predicting traffic volume using machine learning techniques.

The internship helped me understand how real-world problems can be solved using data analysis and predictive modeling. I learned key concepts such as data preprocessing, feature engineering, model building, and evaluation.

I would like to express my sincere gratitude to Upskill Campus, UniConverge Technologies Pvt Ltd (UCT), and my mentors for their guidance and support throughout the internship.

I would like to suggest my juniors and peers to actively participate in internships like this, as they provide practical exposure and improve technical skills.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

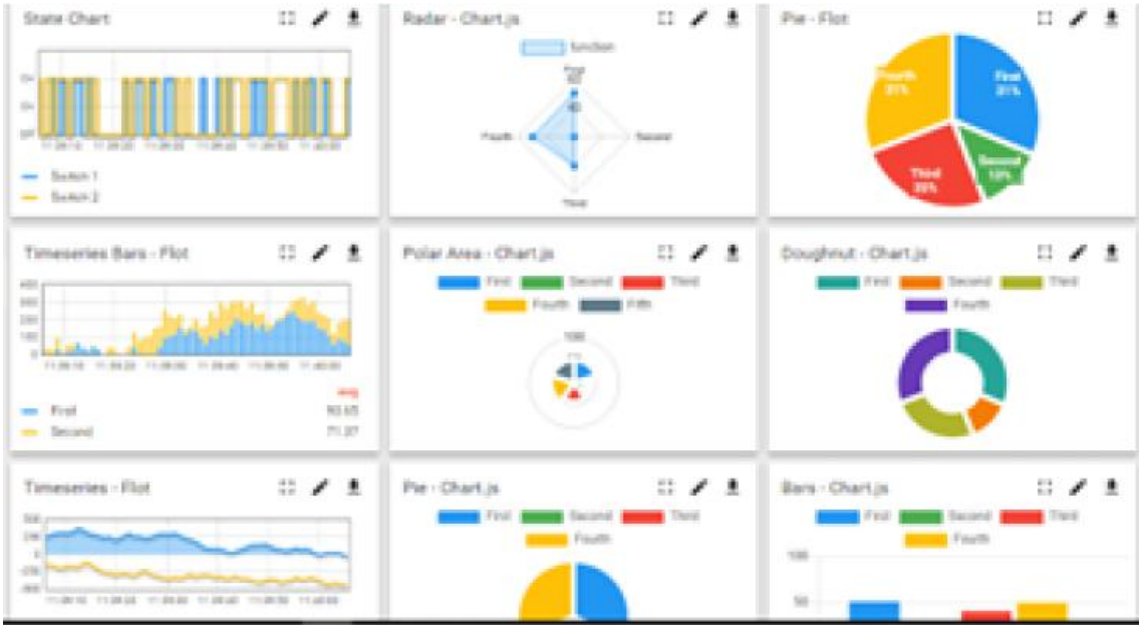
i. UCT IoT Platform (UCT Insight)

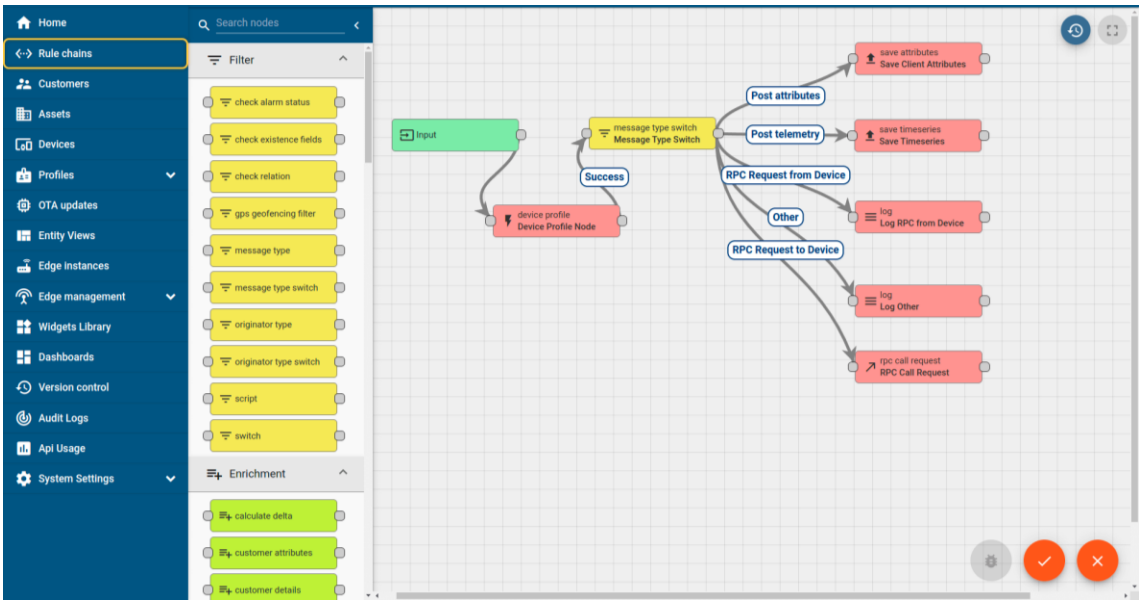
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

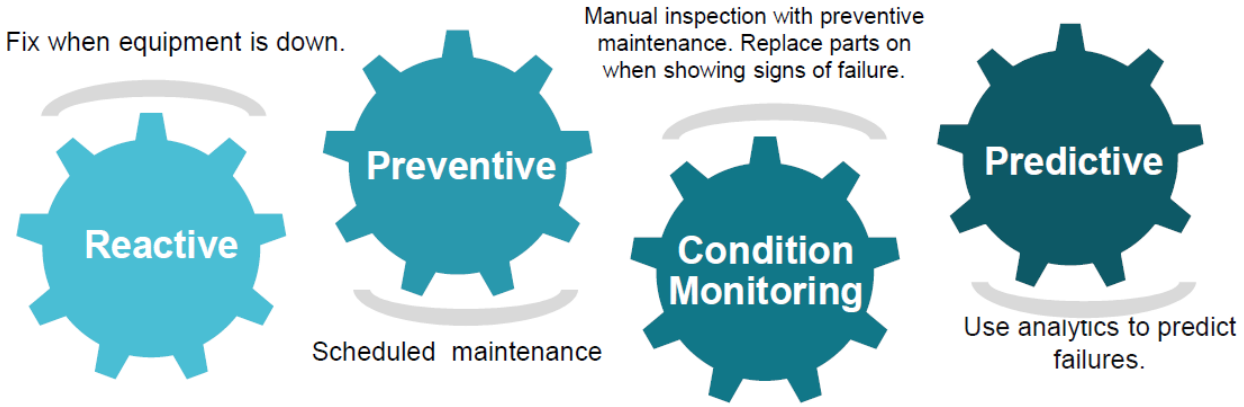
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

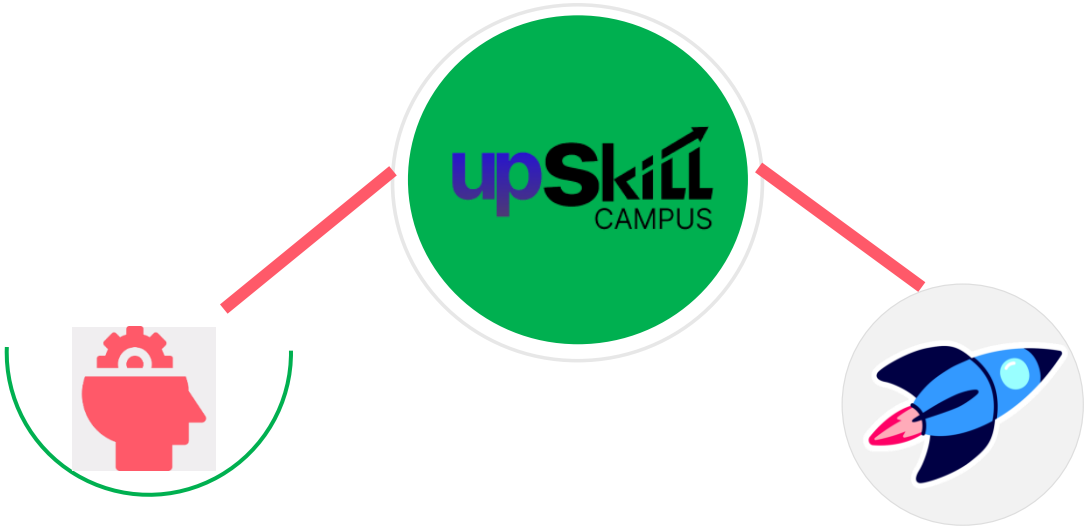
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

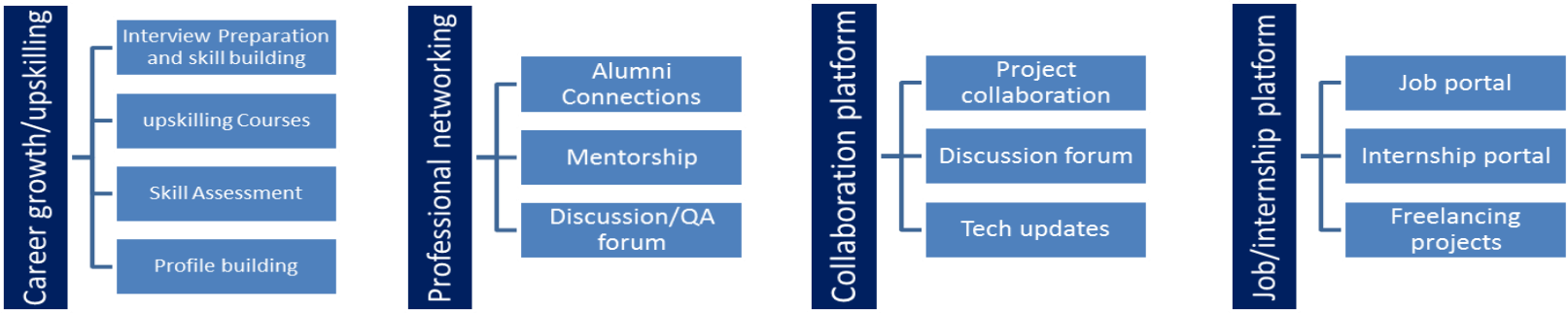
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective of this internship is:

☛ To gain practical experience in Data Science and Machine Learning.

☛ To understand real-world problem-solving using data.

☛ To develop skills in Python programming and machine learning.

☛ To analyze traffic data and predict traffic volume.

☛ To improve problem-solving and analytical thinking skills.

2.5 Reference

[1] https://pandas.pydata.org/

[2] https://scikit-learn.org/

[3] https://matplotlib.org/

2.6 Glossary

Terms	Acronym
Machine Learning	ML
Mean Absolute Error	MAE
Random Forest	RF
Linear Regression	LR
Dataset	Data

3 Problem Statement

The main objective of this project is to predict traffic volume based on time-related features such as hour, day, month, and day of the week.

Accurate traffic prediction helps in better traffic management, reduces congestion, and supports smart city planning. The challenge is to analyze historical data and build a model that can predict future traffic patterns effectively.

4 Existing and Proposed solution

Existing systems often fail to provide accurate traffic predictions due to limited data analysis and lack of advanced techniques.

The proposed solution uses machine learning models such as Linear Regression and Random Forest to analyze historical traffic data. These models help in predicting traffic volume more accurately and efficiently.

This solution improves decision-making for traffic control and helps in better urban planning.

4.1 Code submission (Github link)

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

The proposed system follows a structured machine learning workflow for traffic prediction. Initially, traffic data is collected and preprocessed to remove inconsistencies and extract useful features such as hour, day, month, and day of the week

These features are then used to train machine learning models such as Linear Regression and Random Forest. The trained models are evaluated using appropriate metrics, and the best-performing model is selected for predicting traffic volume.

The system provides an efficient approach to analyze traffic patterns and supports better decision-making for smart city traffic management.

5.1 High Level Diagram of Smart City Traffic Forecasting System

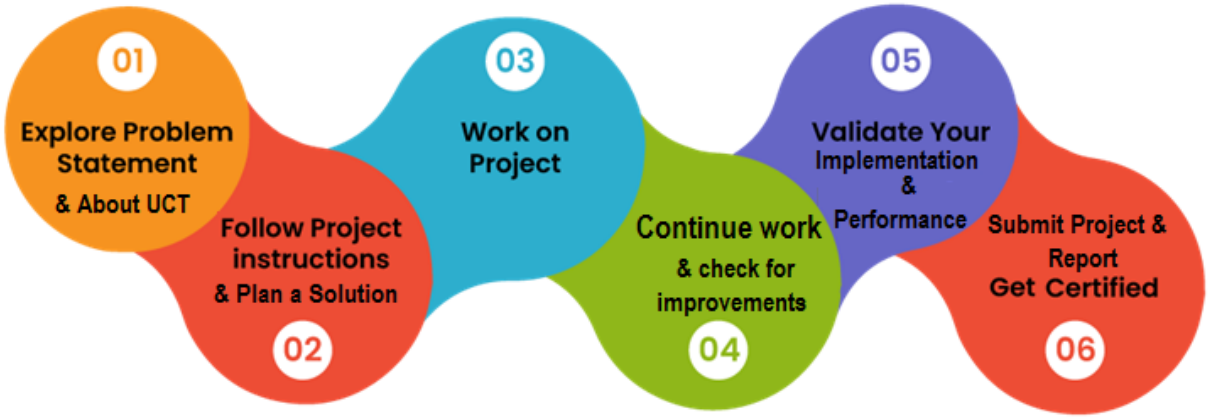


Figure 1: High Level Diagram of Smart City Traffic Forecasting System

5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

The system interface is based on a simple machine learning workflow.

The input to the system is traffic-related data such as hour, day, month, and day of the week. This data is processed and passed to the machine learning model.

The model processes the input data and predicts the traffic volume as output.

The overall flow includes data input, preprocessing, model training, and prediction output. The system does not involve complex hardware or external interfaces, and it operates using Python-based implementation.

6 Performance Test

The performance of the Smart City Traffic Forecasting system was evaluated based on prediction accuracy and model efficiency.

The main constraint considered in this project is accuracy of prediction. Machine learning models were trained and tested on available data to ensure reliable predictions.

Another constraint is computational efficiency. The models used are lightweight and suitable for basic systems without requiring high processing power.

To handle these constraints, proper data preprocessing and feature engineering were performed. Suitable machine learning models such as Linear Regression and Random Forest were selected to balance accuracy and performance.

6.1 Test Plan/ Test Cases

The test plan includes dividing the dataset into training and testing sets.

The models were trained on historical traffic data and tested on unseen data. The main test case was to compare predicted traffic values with actual traffic values to measure accuracy.

6.2 Test Procedure

The dataset was first preprocessed and cleaned. Relevant features such as hour, day, and month were extracted.

The data was split into training and testing datasets. Machine learning models were trained using the training data and predictions were made on the test data.

The results were then evaluated using Mean Absolute Error (MAE).

6.3 Performance Outcome

The performance results showed that the Random Forest model provided better accuracy compared to Linear Regression.

The Mean Absolute Error (MAE) was lower for Random Forest, indicating more accurate predictions.

Overall, the system performs efficiently in predicting traffic volume based on time-related features.

7 My learnings

During this internship, I gained practical knowledge in the field of Data Science and Machine Learning. I learned how to work with datasets, perform data preprocessing, and extract meaningful features from raw data.

I understood how machine learning models such as Linear Regression and Random Forest are used for prediction tasks. I also learned how to evaluate model performance using metrics like Mean Absolute Error (MAE).

This internship helped me improve my Python programming skills, data analysis abilities, and problem-solving approach. It also gave me exposure to real-world applications of machine learning.

Overall, this experience will help me in my career growth by strengthening my technical skills and confidence in working on data-driven projects.

8 Future work scope

In the future, this project can be improved by using real-time traffic data to enhance prediction accuracy. Advanced machine learning and deep learning models can be applied to achieve better performance.

The system can also be integrated with smart city infrastructure to provide real-time traffic monitoring and management solutions. Additional features such as weather data and event-based traffic analysis can be included to make predictions more reliable.

Further improvements can focus on optimizing the model for faster performance and deploying the system as a web-based or mobile application.